

MATH 345 Skeleton Notes

Unit 4: Orthogonality and least squares

Section 4.4: \$QR\$ factorization and least squares

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QR factorization

Recall that a list of vectors is orthonormal when the vectors have length 1 and are pairwise orthogonal. Also recall the notion of an upper triangular matrix from Triangular and Diagonal Matrices.

Definition Let A be an $m \times n$ matrix with independent columns. A

QR factorization of A is an $m \times n$ matrix Q , with orthonormal columns, and an invertible upper triangular $n \times n$ matrix R , with positive entries on its diagonal, so that $A = QR$.

Warning: Rectangular Q In a QR factorization of an $m \times n$ matrix with $m > n$, the matrix Q has orthonormal columns, so $Q^T Q = I_n$. But Q is not square, and QQ^T is not usually I_m .

Normal equations are conceptually important, but QR is often a better computational method. If $A = QR$ with Q having orthonormal columns, then

$$\|Ax - b\| = \|QRx - b\|.$$

Orthogonality reduces the least-squares problem to

$$R\hat{x} = Q^T b,$$

which is triangular and easier to solve accurately.

Activity: Reading QR least-squares code

The following code uses QR to solve the same least-squares problem from the Unit 2 target activity.

```
A = np.array([
    [1.0, 0.0],
    [0.0, 1.0],
    [1.0, 1.0],
])
b = np.array([1.0, 2.0, 4.0])

Q, R = np.linalg.qr(A)
x_qr = np.linalg.solve(R, Q.T @ b)
x_lstsq = np.linalg.lstsq(A, b, rcond=None)[0]

Q.T @ Q, x_qr, x_lstsq
```

1. What should $Q.T @ Q$ be close to?
2. Why does the code solve a system with R instead of a system with A ?
3. Why should x_{qr} and x_{lstsq} agree?
4. Does this mean $Ax = b$ has an exact solution?

The matrix Q has orthonormal columns, so $Q^T Q$ should be close to the identity matrix. The QR factorization rewrites the least-squares problem using $A = QR$, which leads to the triangular system

$$R\hat{\mathbf{x}} = Q^T \mathbf{b}.$$

The vectors $\mathbf{x}_{qr}$ and $\mathbf{x}_{lstsq}$ should agree because they compute the same least-squares solution. This does not mean $A\mathbf{x} = \mathbf{b}$ has an exact solution; it means the fitted vector $A\hat{\mathbf{x}}$ is the closest reachable output.

Activity

Let

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}.$$

Check that the columns are orthonormal and compute $Q^T Q$.

Column 1: $\|\mathbf{q}_1\|^2 = \left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$

Column 2: $\|\mathbf{q}_2\|^2 = \left(\frac{1}{\sqrt{6}}\right)^2 + \left(-\frac{2}{\sqrt{6}}\right)^2 + \left(\frac{1}{\sqrt{6}}\right)^2 = 1$

Orthogonality:

$$\begin{aligned}\mathbf{q}_1 \cdot \mathbf{q}_2 &= \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{6}} + \frac{1}{\sqrt{3}} \cdot \left(-\frac{2}{\sqrt{6}}\right) + \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{6}} \\ &= \frac{1}{\sqrt{18}} - \frac{2}{\sqrt{18}} + \frac{1}{\sqrt{18}} = 0\end{aligned}$$

Compute $Q^T Q$:

$$\begin{aligned}Q^T Q &= \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{3} + \frac{1}{3} + \frac{1}{3} & \frac{1}{\sqrt{18}} - \frac{2}{\sqrt{18}} + \frac{1}{\sqrt{18}} \\ \frac{1}{\sqrt{18}} - \frac{2}{\sqrt{18}} + \frac{1}{\sqrt{18}} & \frac{1}{6} + \frac{4}{6} + \frac{1}{6} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2\end{aligned}$$

Activity

Solve the linear system $R\mathbf{x} = \mathbf{b}$ where

$$R = \begin{bmatrix} 4 & 2 & 7 \\ 0 & 3 & 5 \\ 0 & 0 & 2 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 30 \\ 21 \\ 6 \end{bmatrix}$$

For an upper triangular matrix like R , we can solve the system using back-substitution. Starting with the last equation:

$$2x_3 = 6$$

$$x_3 = 3$$

Now substitute this value into the second equation:

$$3x_2 + 5x_3 = 21$$

$$3x_2 + 5(3) = 21$$

$$3x_2 + 15 = 21$$

$$3x_2 = 6$$

$$x_2 = 2$$

Finally, substitute both known values into the first equation:

$$\begin{aligned}4x_1 + 2x_2 + 7x_3 &= 30 \\4x_1 + 2(2) + 7(3) &= 30 \\4x_1 + 4 + 21 &= 30 \\4x_1 &= 5 \\x_1 &= 1.25\end{aligned}$$

The solution is $\mathbf{x} = (1.25, 2, 3)$.

The QR factorization is a matrix version of the *Gram-Schmidt orthogonalization process*, which has many applications in many numerical algorithms.

Theorem Every $m \times n$ matrix A with linearly independent columns has a QR factorization $A = QR$.

Proof Let $A = [\mathbf{c}_1 \mathbf{c}_2 \cdots \mathbf{c}_n]$ be the matrix with columns $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$. Apply Gram-Schmidt process to obtain orthogonal vectors $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n$ where: $\mathbf{f}_1 = \mathbf{c}_1$ and

$$\mathbf{f}_k = \mathbf{c}_k - \frac{\mathbf{c}_k \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1 - \frac{\mathbf{c}_k \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} \mathbf{f}_2 - \dots - \frac{\mathbf{c}_k \cdot \mathbf{f}_{k-1}}{\|\mathbf{f}_{k-1}\|^2} \mathbf{f}_{k-1}$$

for $k = 2, 3, \dots, n$.

Let $\mathbf{q}_k = \frac{\mathbf{f}_k}{\|\mathbf{f}_k\|}$ for each k . Then $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n$ are orthonormal vectors.

We can rewrite each column $\mathbf{c}_k$ as a linear combination of the $\mathbf{q}_i$: $\mathbf{c}_1 =$

$$\|\mathbf{f}_1\|\mathbf{q}_1,$$

$$\mathbf{c}_2 = (\mathbf{c}_2 \cdot \mathbf{q}_1)\mathbf{q}_1 + \|\mathbf{f}_2\|\mathbf{q}_2$$

$$\mathbf{c}_3 = (\mathbf{c}_3 \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{c}_3 \cdot \mathbf{q}_2)\mathbf{q}_2 + \|\mathbf{f}_3\|\mathbf{q}_3$$

and so on. This gives us the matrix factorization:

$$\begin{aligned} A &= [\mathbf{c}_1 \mathbf{c}_2 \cdots \mathbf{c}_n] \\ &= [\mathbf{q}_1 \mathbf{q}_2 \cdots \mathbf{q}_n] \begin{bmatrix} \|\mathbf{f}_1\| & \mathbf{c}_2 \cdot \mathbf{q}_1 & \mathbf{c}_3 \cdot \mathbf{q}_1 & \cdots & \mathbf{c}_n \cdot \mathbf{q}_1 \\ 0 & \|\mathbf{f}_2\| & \mathbf{c}_3 \cdot \mathbf{q}_2 & \cdots & \mathbf{c}_n \cdot \mathbf{q}_2 \\ 0 & 0 & \|\mathbf{f}_3\| & \cdots & \mathbf{c}_n \cdot \mathbf{q}_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \|\mathbf{f}_n\| \end{bmatrix} \end{aligned}$$

So $A = QR$ where $Q = [\mathbf{q}_1 \mathbf{q}_2 \cdots \mathbf{q}_n]$ has orthonormal columns and R is upper triangular with positive diagonal entries.

Activity

Consider the matrix

$$A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}.$$

Find its QR factorization.

The columns of A are $\mathbf{c}_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\mathbf{c}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Set $\mathbf{f}_1 = \mathbf{c}_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Then $\|\mathbf{f}_1\| = \sqrt{3^2 + 4^2} = 5$.

Compute $\mathbf{q}_1 = \frac{\mathbf{f}_1}{\|\mathbf{f}_1\|} = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$.

Compute $\mathbf{f}_2 = \mathbf{c}_2 - \frac{\mathbf{c}_2 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} \mathbf{f}_1$:

$$\mathbf{c}_2 \cdot \mathbf{f}_1 = 1 \cdot 3 + 2 \cdot 4 = 11$$

$$\mathbf{f}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \frac{11}{25} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 - \frac{33}{25} \\ 2 - \frac{44}{25} \end{bmatrix} = \begin{bmatrix} \frac{25-33}{25} \\ \frac{50-44}{25} \end{bmatrix} = \begin{bmatrix} -\frac{8}{25} \\ \frac{6}{25} \end{bmatrix}$$

$$\text{Compute } \|\mathbf{f}_2\| = \sqrt{\left(-\frac{8}{25}\right)^2 + \left(\frac{6}{25}\right)^2} = \sqrt{\frac{64+36}{625}} = \sqrt{\frac{100}{625}} = \frac{10}{25} = \frac{2}{5}.$$

$$\text{Compute } \mathbf{q}_2 = \frac{\mathbf{f}_2}{\|\mathbf{f}_2\|} = \frac{1}{\frac{2}{5}} \begin{bmatrix} -\frac{8}{25} \\ \frac{6}{25} \end{bmatrix} = \frac{5}{2} \begin{bmatrix} -\frac{8}{25} \\ \frac{6}{25} \end{bmatrix} = \begin{bmatrix} -\frac{4}{5} \\ \frac{3}{5} \end{bmatrix}.$$

Calculate the entries of R :

$$r_{11} = \|\mathbf{f}_1\| = 5$$

$$r_{12} = \mathbf{c}_2 \cdot \mathbf{q}_1 = 1 \cdot \frac{3}{5} + 2 \cdot \frac{4}{5} = \frac{3+8}{5} = \frac{11}{5}$$

$$r_{22} = \|\mathbf{f}_2\| = \frac{2}{5}$$

Therefore, $A = QR$ where:

$$Q = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix} \quad R = \begin{bmatrix} 5 & \frac{11}{5} \\ 0 & \frac{2}{5} \end{bmatrix}$$

Activity

Find the QR factorization of the matrix

$$A = \begin{bmatrix} 6 & 2 \\ 0 & 3 \end{bmatrix}.$$

The columns of A are $\mathbf{c}_1 = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$ and $\mathbf{c}_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

$$\mathbf{f}_1 = \mathbf{c}_1 = \begin{bmatrix} 6 \\ 0 \end{bmatrix}, \|\mathbf{f}_1\| = 6$$

$$\mathbf{q}_1 = \frac{\mathbf{f}_1}{\|\mathbf{f}_1\|} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\mathbf{c}_2 \cdot \mathbf{q}_1 = 2 \cdot 1 + 3 \cdot 0 = 2$$

$$\mathbf{f}_2 = \mathbf{c}_2 - (\mathbf{c}_2 \cdot \mathbf{q}_1)\mathbf{q}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$\|\mathbf{f}_2\| = 3$$

$$\mathbf{q}_2 = \frac{\mathbf{f}_2}{\|\mathbf{f}_2\|} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore, $A = QR$ where:

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad R = \begin{bmatrix} 6 & 2 \\ 0 & 3 \end{bmatrix}$$

Theorem Every square, invertible matrix A can be written as $A = QR$, where Q is an orthogonal matrix, and R is an upper triangular matrix with positive diagonal entries.